

Hybridisation & orbitals

no. of regions of electron density about C			
shape about C			
hybridisation of C			
illustrate the concept of orbital mixing			
draw and label the hybridised orbitals of C			
label the hybridised + unhybridised orbitals of C			
draw a labelled diagram to show the orbital involved in bonds formed between the two C atoms			

Effect of hybridisation on bond strength & bond length

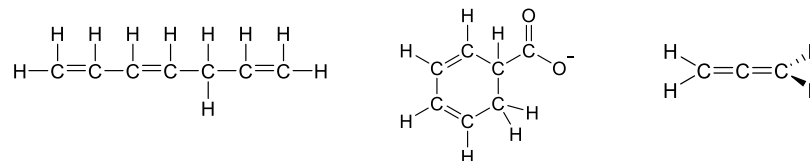
compounds			
	ethane	ethene	ethyne
orbital overlap for C—H bond	$sp^3 - s$		
% s-character of hybridised orbitals in C	25%		

- Explain which compound has the strongest C—H bond strength and shortest C—H bond length.
- the _____ orbital of C in ethyne has the greatest _____ and is the _____ diffuse
 - therefore the $sp - s$ orbital overlap is the most _____
 - hence the C—H bond in ethyne is the strongest and its bond length the shortest

Delocalisation of π electrons

- State the condition for delocalisation of π electrons (i.e. can move about freely across the p orbitals)

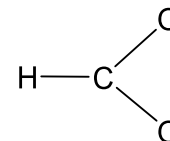
- Circle the atoms where electrons can be delocalised



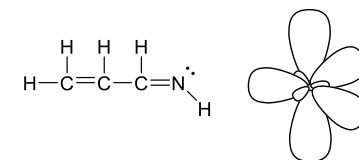
Note: To identify atoms with p orbitals...

- check for atoms with : _____ / _____ / _____

- By drawing the orbitals involved, show how delocalisation of e^- occurs in HCOO^-



- The N atom in the molecule is sp^2 hybridised. Draw the 5 valence e^- of N in its valence orbitals

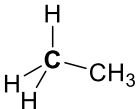
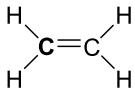
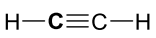
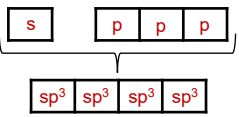
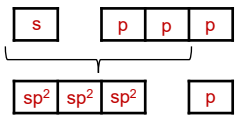
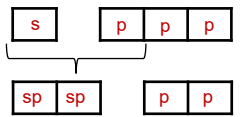
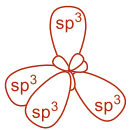
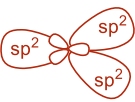
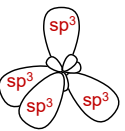
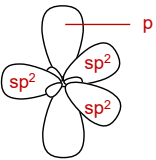
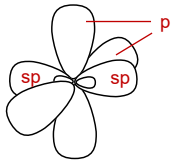
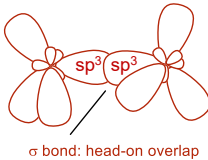
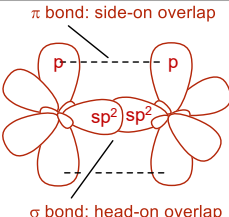
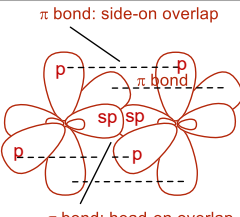


- The lone pair of e^- on N is unable to delocalise into the π system. Explain why this is so.

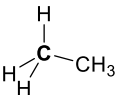
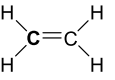
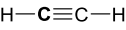
The lone pair of e^- of N are in the _____ orbital. The sp^2 orbital of N _____ continuously with the p orbitals of the C atoms. Hence, the lone pair of e^- on N cannot delocalise.

- Number of delocalised e^- in HCOO^- = _____

Hybridisation & orbitals

			
no. of regions of electron density about C	4	3	2
shape about C	tetrahedral	trigonal planar	linear
hybridisation of C	sp^3	sp^2	sp
illustrate the concept of orbital mixing			
draw and label the hybridised orbitals of C			
label the hybridised + unhybridised orbitals of C			
draw a labelled diagram to show the orbital involved in bonds formed between the two C atoms	 σ bond: head-on overlap	 π bond: side-on overlap σ bond: head-on overlap	 π bond: side-on overlap σ bond: head-on overlap

Effect of hybridisation on bond strength & bond length

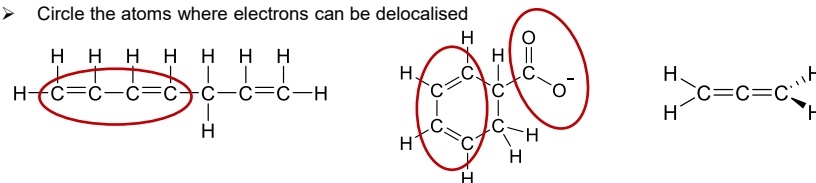
compounds	 ethane	 ethene	 ethyne
orbital overlap for C—H bond	$sp^3 - s$	$sp^2 - s$	$sp - s$
% s-character of hybridised orbitals in C	25%	33%	50%

- Explain which compound has the strongest C—H bond strength and shortest C—H bond length.
- the sp orbital of C in ethyne has the greatest s-character and is the least diffuse
 - therefore the $sp - s$ orbital overlap is the most effective
 - hence the C—H bond in ethyne is the strongest and its bond length the shortest

Delocalisation of π electrons

- State the condition for delocalisation of π electrons (i.e. can move about freely across the p orbitals)
- continuous side-on overlap of p-orbitals over at least 3 adjacent atoms

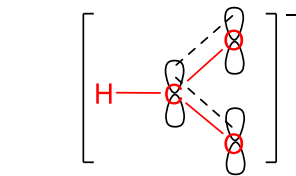
- Circle the atoms where electrons can be delocalised



Note: To identify atoms with p orbitals...

- check for atoms with: π bond / l.p. of e^- / incomplete octet

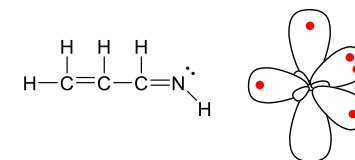
- By drawing the orbitals involved, show how delocalisation of e^- occurs in HCOO^-



p-orbitals of O, C and O atoms overlap continuously

- Number of delocalised e^- in $\text{HCOO}^- = 4$

- The N atom in the molecule is sp^2 hybridised. Draw the 5 valence e^- of N in its valence orbitals



- The lone pair of e^- on N is unable to delocalise into the π system. Explain why is this so.

The lone pair of e^- of N are in the sp^2 orbital. The sp^2 orbital of N does not overlap continuously with the p orbitals of the C atoms. Hence, the lone pair of e^- on N cannot delocalise.